$f(x) = e^x$
$\ln(1+x)$

Tyler Series:

① ↳ Polynomial approximation.

$P(x) = a_0 + a_1 x + a_2 x^2 + \cdots$

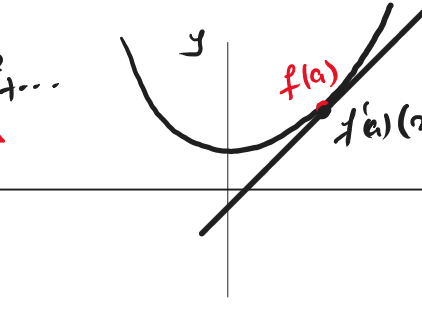
$P(x) = 2x + 3$

② ↳ Derivative:

$P(x) = f(x)$
$P'(x) = f'(x)$
$P''(x) = f''(x)$
$\vdots$

$P(x) = f(x)$
$x = a$
$P(a) = f(a)$. at centre point 'a'.
$x = 0$
$P(0) = f(0)$, Maclaurin Series, at centre point 0.

$$y = \underbrace{f(a)}_{\text{Point Value}} + \underbrace{f'(a)(x-a)}_{\text{Tangent Linear}} + \underbrace{\frac{f''(a)}{2!}(x-a)^2}_{\text{Quadrative Curve.}} + \cdots$$



Tyler Series: $(x=a)$

$$f(x) = f(a) + \frac{f'(a)(x-a)}{1!} + \frac{f''(a)(x-a)^2}{2!} + \frac{f'''(a)(x-a)^3}{3!} + \cdots$$

Why factorial:

$P(x) = A + B(x-a) + C(x-a)^2 + D(x-a)^{\cancel{1}3} + \cdots$

$P'(x) = B + 2C(x-a) + \cdots$

$P''(x) = 2C$

$\frac{P''(x)}{2} = \frac{2C}{2}$

$\frac{P''(x)}{2!} = C$

$\frac{P'''(x)}{3!} = D$

$$P(x) = A + B(x-a) + \frac{P''(x)}{2!}(x-a)^2 + \frac{P'''(x)}{3!}(x-a)^3 + \cdots$$

General Tyler Series: $(x=a)$

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

Example: $f(x) = e^x$

$f'(x) = e^x\left(\frac{d}{dx}x\right) = e^x(1) = e^x$

$f''(x) = e^x$

$\vdots$

At $x = 0 = a$

$f(0) = e^0 = 1$
$f'(0) = e^0 = 1$
$f''(0) = e^0 = 1 \ldots$

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(x)(x-a)^2}{2!} + \cdots$$

$e^x = 1 + x + \frac{1}{2!} \cdot x^2 + \cdots$

$e^x = 1 + x + \frac{x^2}{2!} + \cdots$ Approx. //

Let e^3 upto $f''(x)$

$e^3 = 1 + 3 + \frac{(3)^2}{2 \times 1}$

$= 1 + 3 + \frac{9}{2}$

$= \frac{2+6+9}{2} = \frac{17}{2} = 8.5$

$e^3 \approx 8.5$ //.

② $f(x) = \sin x$.

$f'(x) = \cos x\left(\frac{d}{dx}x\right) = \cos x(1) = \cos x$

$f''(x) = -\sin x$

$f'''(x) = -\cos x$

$f^{4}(x) = \sin x$

at $x = 0$.

$\sin 0 = 0$
$\cos 0 = 1$

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \cdots$$

$$\sin x = 0 + 1(x) + \frac{0}{2!}(x)^2 + \frac{(+1)}{3!}(x)^3 + \frac{0}{4!}(x)^4 + \frac{(-1)}{5!}(x)^5 + \cdots$$

$$\sin x = x^1 - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} + \cdots$$ Odd indexes

③ $\cos x$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$$ Even Indexes.

Approximate $\approx$

Structural Eng: $EI\frac{d^4 y}{dx^4} = w(x)$

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!}f''(x) + \cdots$$

$$f(x-h) = f(x) - hf'(x) + \frac{h^2}{2!}f''(x) + \cdots$$

Subtract above two Eq:

$$f(x+h) - f(x-h) = 2hf'(x) + \cdots$$

$$\frac{f(x+h) - f(x-h)}{2h} = f'(x)$$ //

Imp: Error: is Second derivative

$$\mathcal{E} = \frac{f''(x)}{2!}(x-a)^2$$

$$R_1(x) = \frac{f''(\xi)}{2!}(x-a)^2$$ //.

Tyler Polynomial = Finite
Tyler Series = Infinte

Q2 Coefficient of x^5 in $\sin(\underline{2x})$

$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$ General TS.

Put $x = 2x$

$$\sin(2x) = 2x - \frac{(2x)^3}{3!} + \underline{\underline{\frac{(2x)^5}{5!}}} - \cdots$$

Coefficient of $x^5 = \frac{(2x)^5}{5!}$

$= \frac{2^5 x^5}{5 \times 4 \times 3 \times 2 \times 1}$

$= \frac{32}{120}x^5$

$= \frac{\cancel{32}\,4}{\cancel{120}\,15}$

$= 4/15$ //.